

Run report

2026-09-11T005621Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	1,025,884,356	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	4.392×10^{-4}	best_epoch	92
compile_cache_warm	True		
variables	m [GeV], $\ln \rho$, $\lambda_{0.5}^1$, w , λ_2^1 , z_g , $\tau_{21}^{(\beta=1)}$, M , n_{ch} , f_{ch} , p_T^D , q		
mmd_sigmas_detector	$3.107 \times (1/2, 1/\sqrt{2}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (1/2, 1/\sqrt{2}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	36.21	73.5	train
transfer	2.037		train
compile	3.415		train
epochs	29.52		train
select	1.043		train
particle_mmd	0.4087	0.829	train
save	0.5197	1.05	train
data	1.506	3.06	cache hit – load
load	2.463	5.00	load
plots	8.164	16.6	load
evaluate	0.005268	0.0107	load
total	49.28	100	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	208.9	138.8	33.6	12.09	94.2
$\ln \rho$	227.9	20.71	90.9	17.03	92.5
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	262.6	12.84	95.1	13.27	94.9
w	265.6	9.978	96.2	13.04	95.1
λ_2^1	255.4	9.215	96.4	15.38	94.0
Splitting and 2-prong substructure					
$z_g^\dagger$	60.45	60.45	0.0	9.894	83.6
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	19.30	84.2
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	224.2	177.3	20.9	23.98	89.3
n_{ch}	220.8	172.3	22.0	27.88	87.4
f_{ch}	52.86	40.94	22.6	8.286	84.3
p_T^D	256.8	30.80	88.0	19.83	92.3
q	79.67	17.51	78.0	8.517	89.3

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	6.762	3.666	45.8	0.09497	98.6
$\ln \rho$	8.383	0.4017	95.2	0.1613	98.1
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	9.359	0.4714	95.0	0.1216	98.7
w	9.139	0.4527	95.0	0.1202	98.7
λ_2^1	7.776	0.1728	97.8	0.1299	98.3
Splitting and 2-prong substructure					
$z_g^\dagger$	0.6723	0.6723	0.0	0.09017	86.6
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.1877	94.4
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	9.640	7.632	20.8	0.1896	98.0
n_{ch}	7.512	6.314	16.0	0.1783	97.6
f_{ch}	1.033	0.7960	22.9	0.09241	91.1
p_T^D	8.223	1.311	84.1	0.1624	98.0
q	1.576	0.2512	84.1	0.08549	94.6

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

VLC divergence [$\times 10^3$]

Observable	Sim.	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	26.93	14.62	45.7	0.3611	98.7
$\ln \rho$	33.31	1.576	95.3	0.6251	98.1
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	37.19	1.877	95.0	0.4802	98.7
w	36.32	1.800	95.0	0.4728	98.7
λ_2^1	30.90	0.6583	97.9	0.4816	98.4
Splitting and 2-prong substructure					
$z_g^\dagger$	2.683	2.683	0.0	0.3606	86.6
$\tau_{21}^{(\beta=1)}$	13.44	7.712	42.6	0.7459	94.5
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	38.25	30.31	20.8	0.7265	98.1
n_{ch}	29.81	25.07	15.9	0.6859	97.7
f_{ch}	4.125	3.177	23.0	0.3696	91.0
p_T^D	32.65	5.211	84.0	0.6446	98.0
q	6.261	0.9940	84.1	0.3316	94.7

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	111.7	94.08	15.8	89.70	19.7
$\ln \rho$	253.1	32.57	87.1	8.667	96.6
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	303.0	18.71	93.8	10.58	96.5
w	284.1	12.29	95.7	8.025	97.2
λ_2^1	253.8	11.13	95.6	7.723	97.0
Splitting and 2-prong substructure					
$z_g^\dagger$	113.4	113.4	0.0	10.22	91.0
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	14.48	94.4
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	256.8	204.8	20.2	20.47	92.0
n_{ch}	234.6	188.6	19.6	19.86	91.5
f_{ch}	45.78	45.33	1.0	5.725	87.5
p_T^D	236.3	55.36	76.6	18.28	92.3
q	79.37	18.15	77.1	9.600	87.9

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JS divergence [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	1.963	11.86	−504.2	1.507	23.3
$\ln \rho$	9.878	1.731	82.5	0.1468	98.5
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	12.34	2.009	83.7	0.1586	98.7
w	11.39	1.465	87.1	0.1622	98.6
λ_2^1	8.824	0.8335	90.6	0.1231	98.6
Splitting and 2-prong substructure					
$z_g^\dagger$	2.041	2.041	0.0	0.1553	92.4
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.1662	98.4
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	7.842	12.83	−63.7	0.2435	96.9
n_{ch}	6.447	9.773	−51.6	0.1745	97.3
f_{ch}	1.025	0.9435	7.9	0.1111	89.2
p_T^D	8.342	4.029	51.7	0.2929	96.5
q	2.027	0.9958	50.9	0.1093	94.6

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

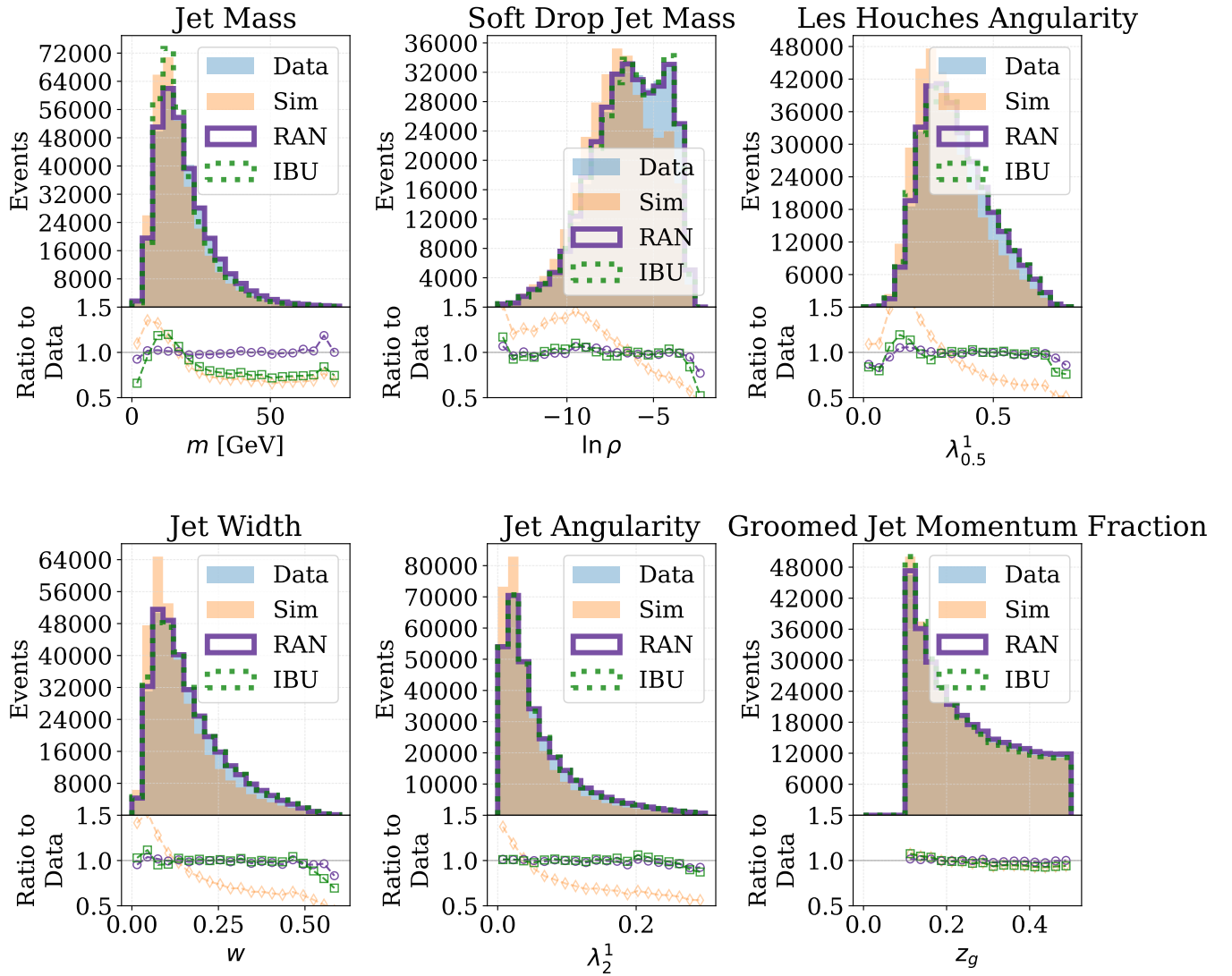
VLC divergence [$\times 10^3$]

Observable	Gen	IBU	IBU impr. (%)	RAN	RAN impr. (%)
Mass and hard scale (IRC-safe kinematics)					
m [GeV]	7.809	45.93	−488.2	5.986	23.3
$\ln \rho$	39.27	6.889	82.5	0.5715	98.5
Continuous angularities (IRC-safe jet shapes)					
$\lambda_{0.5}^1$	48.98	8.014	83.6	0.6288	98.7
w	45.23	5.837	87.1	0.6332	98.6
λ_2^1	35.08	3.315	90.5	0.4763	98.6
Splitting and 2-prong substructure					
$z_g^\dagger$	8.086	8.086	0.0	0.6195	92.3
$\tau_{21}^{(\beta=1)}$	41.75	26.17	37.3	0.6576	98.4
Hadronization, multiplicity and fragmentation (IRC-unsafe)					
M	31.10	49.85	−60.3	0.9307	97.0
n_{ch}	25.58	38.38	−50.1	0.6609	97.4
f_{ch}	4.095	3.766	8.0	0.4440	89.2
p_T^D	33.05	16.05	51.4	1.156	96.5
q	7.982	3.921	50.9	0.3862	95.2

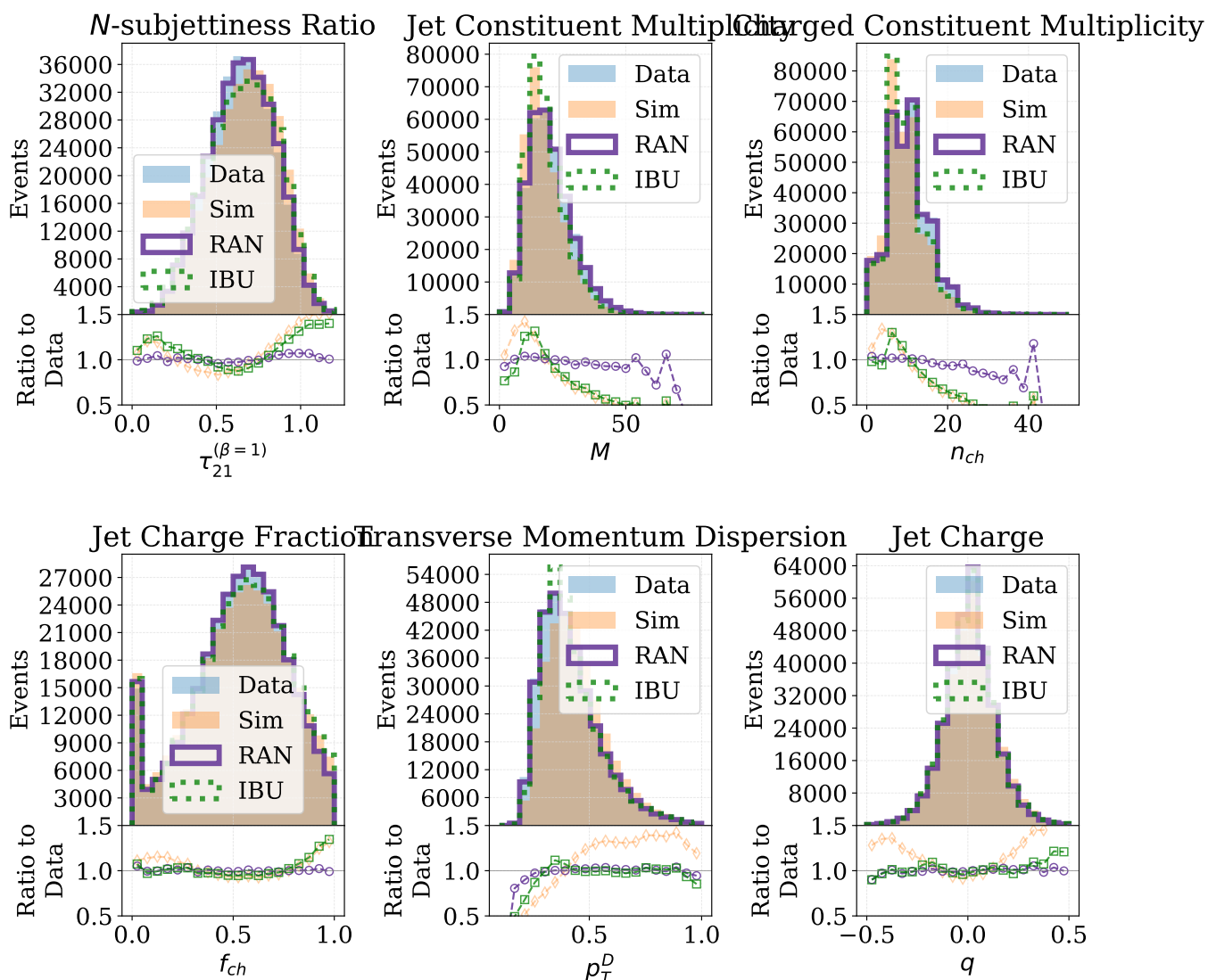
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Detector-level distributions

Detector Level (1 of 2)

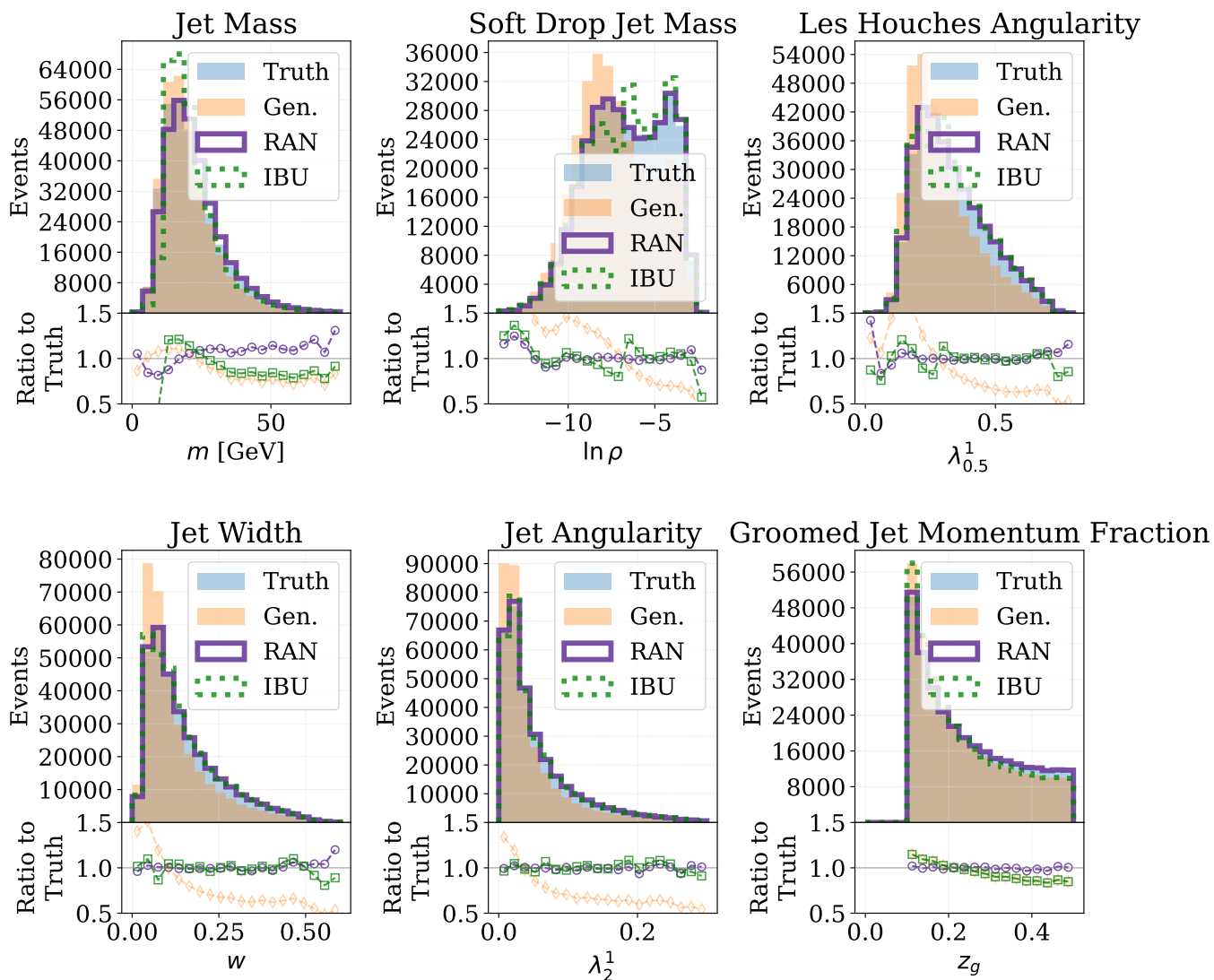


Detector Level (2 of 2)

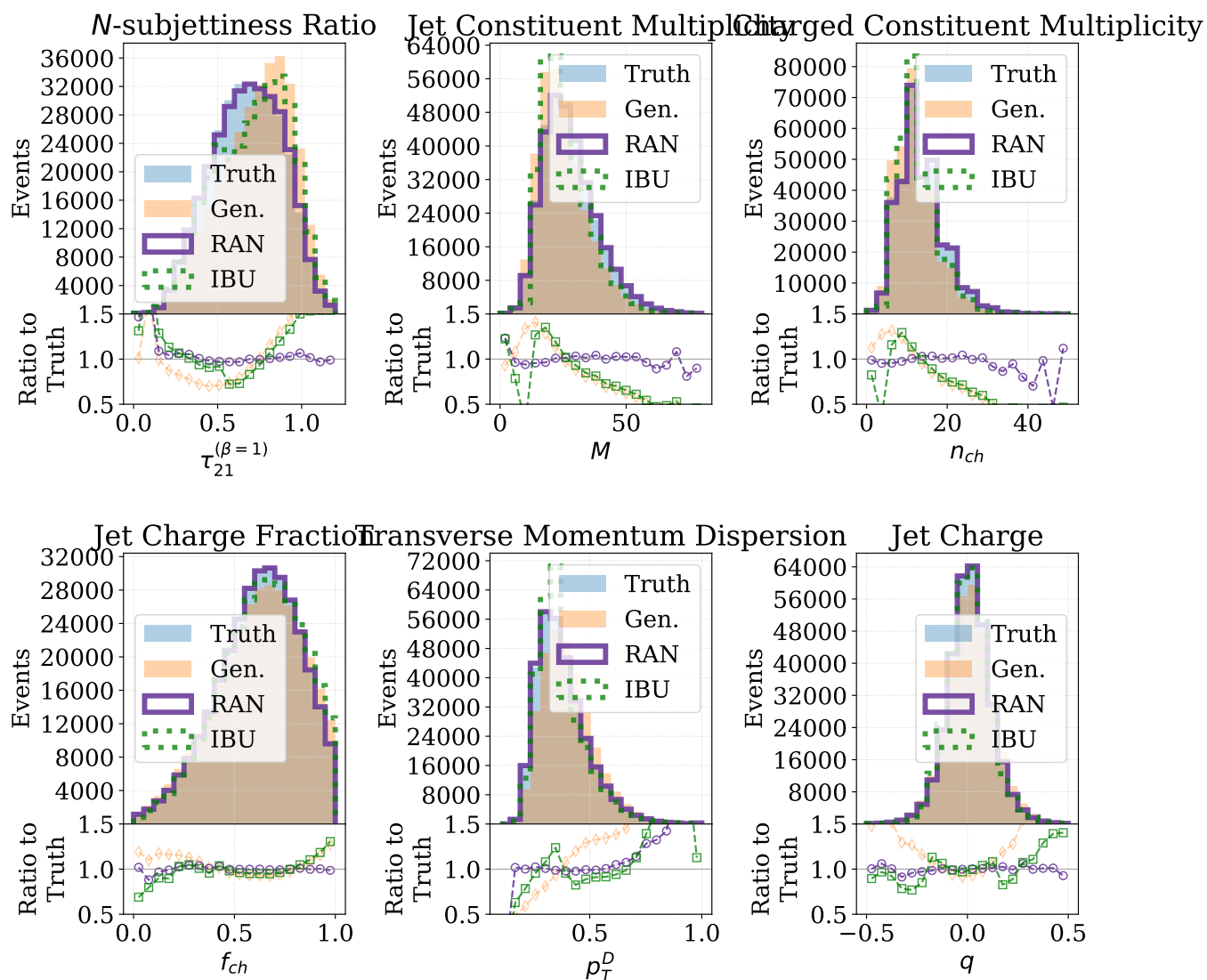


Particle-level distributions

Particle Level (1 of 2)

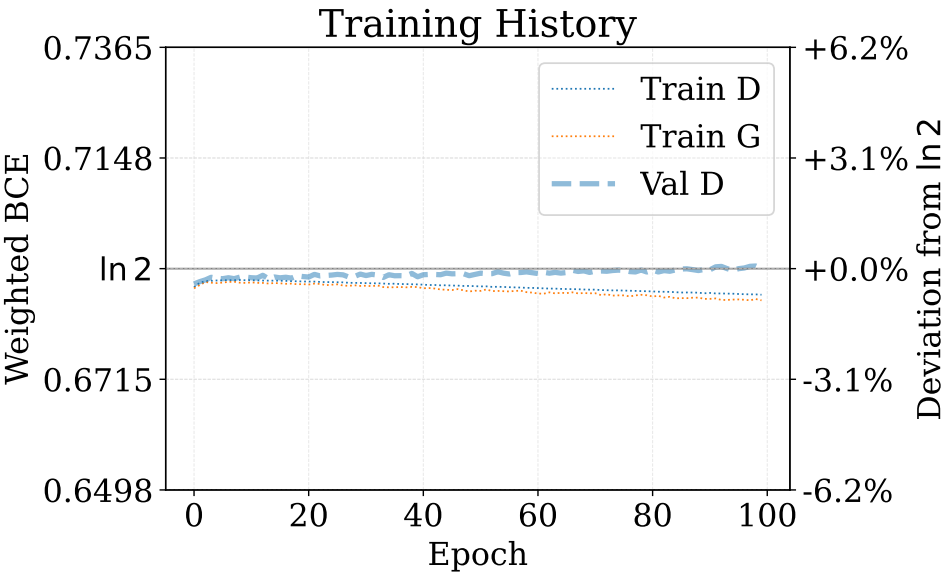


Particle Level (2 of 2)



Training diagnostics

Training losses



Selection criteria

